

A Lightweight Audio-Driven Blendshape Prediction Pipeline for Agent's Facial Animation in XR

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ABSTRACT

Real-time, expressive facial animation is essential for social presence in Extended Reality (XR) and autonomous virtual agents, yet existing solutions typically trade emotional fidelity for efficiency. We present a speech-driven 3D facial animation framework for standalone XR deployment. We utilize acoustic representations coupled with a Bidirectional LSTM and Multi-Head Attention architecture to map continuous audio streams to high-fidelity facial blendshapes. Trained on the emotional MEAD dataset, the model demonstrates strong generalization with a test MSE of 0.0088, particularly in high-intensity emotions. We further validate the pipeline by deploying it into Unity for future interactive virtual agents.

Index Terms: Audio-driven facial animation, Digital virtual agents, Facial blendshapes, Extended reality.

1 INTRODUCTION

Extended reality (XR) is reshaping human computer interaction and enabling new forms of embodied social communication. The success of these platforms depends heavily on achieving social presence, which is the psychological sense of being co-located with others in a shared virtual space. A major challenge in realizing this presence is the real-time generation of expressive facial animation for digital virtual agents. Existing solutions for animating virtual faces fall short, forcing a compromise between performance and fidelity. Traditional game-engine techniques rely on pre-scripted animations or viseme-based lip-sync, which are computationally efficient but fail to capture co-articulation or emotional prosody, resulting in robotic behavior. On the other hand, recent generative-AI talking-head systems achieve near-photorealistic quality but require offline rendering or powerful server-side hardware, making them unsuitable for standalone XR headsets. Audio-driven facial animation has emerged as a promising alternative, transforming speech signals into dynamic 3D facial motion. However, the majority of existing work does not address the broader pipeline needed for fully autonomous, real-time facial animation in immersive environments.

Our work represents an initial step toward a larger vision to have a modular framework for facial animation generation in XR. In this broader framework, user performance data (e.g., speech, intent, or inferred affect) can be transformed into audio, enriched transcription, or emotion-annotated signals using large language mod-

els. These signals then drive lightweight, real-time facial animation models optimized for XR deployment. In this paper, we focus on a neural architecture that maps raw speech features to facial blendshape coefficients in real time, capturing both phonetic accuracy and emotional intensity.

2 RELATED WORK

Research on audio-driven facial animation has progressed from rule-based systems to modern deep learning methods, with a central goal of producing natural, emotionally coherent motion from audio alone. Early neural approaches primarily mapped phonetic cues to facial parameters. MeshTalk [5] introduced a cross-modal disentanglement strategy that separates speech-correlated movements (such as lip motion) from unrelated behaviors like blinking. This categorical latent space reduces the uncanny valley effect but requires complex training procedures and does not inherently guarantee stable emotional coherence. Transformer-based models later became dominant for modeling long-range temporal structure. FaceFormer [3] used an autoregressive Transformer with cross-modal attention to generate 3D facial motion and achieved strong lip-sync accuracy. However, its sequential inference results in higher computational cost, making such architectures difficult to deploy in low-latency XR settings. Beyond phonetic accuracy, recent work emphasizes emotional expressiveness. EMOTE [2] enables explicit emotional control through a content-emotion exchange mechanism. Although these methods improve expressiveness, they depend on explicit emotion labels, which are unavailable in spontaneous interaction. As a result, they struggle to capture natural, unscripted emotional cues like laughter, sighs, or subtle affective shifts that arise organically in real conversation. Meanwhile, efforts in real-time facial animation focus more on rendering efficiency than audio-driven synthesis. Recent stylized-animation systems [4] showed that machine learning can outperform commercial tools in expressiveness, but these pipelines rely on image inputs rather than continuous speech.

Existing work reveals a gap between audio-driven high-fidelity emotional facial animation systems and real-time solutions in XR. Our work aims to bridge this divide by combining emotion-aware acoustic features with a lightweight architecture for continuous audio streams in standalone XR environments, achieving high-fidelity animation under strict real-time latency constraints.

3 METHODOLOGY

Dataset: We utilized the Multi-view Emotional Audio-visual Dataset (MEAD) [6] for training and evaluation. MEAD is a large-scale dataset designed to address the scarcity of high-quality emotional talking-face data. The dataset spans eight distinct emotion categories (neutral, angry, disgust, contempt, fear, happy, sad, surprised) across three intensity levels. From this set, we selected a subset comprising six actors (M005, M009, M011, W016, W028, W036). To ensure optimal facial feature visibility, we restricted data to the front camera angle.

Feature Extraction and Preprocessing: We employed Mediapipe Face Landmarker to process the video data. For each frame,

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This work was supported in part by the ITRC Support Program under Grant IITP-2026-RS-2022 00156354 and in part by the Metaverse Support Program to Nurture the Best Talents under Grant IITP-2026-RS-2023-00254529 funded by the Ministry of Science and ICT of Korea and the Institute of Information and Communications Technology Planning and Evaluation (IITP).

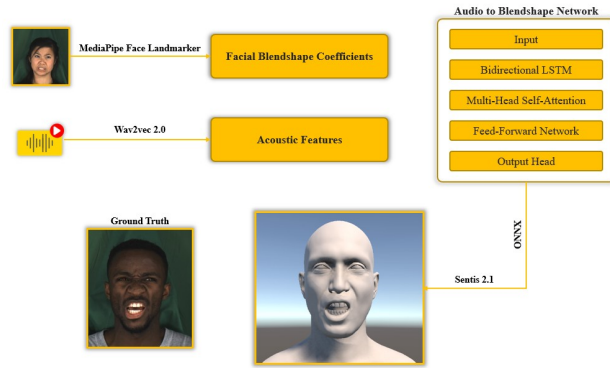


Figure 1: Overview of the proposed pipeline, consisting of audio and visual feature extraction, BiLSTM + multi-head attention, and deployment and visualization in Unity.

the system extracted 52 facial blendshape coefficients, representing the target facial muscle activations. To obtain robust acoustic features, we utilized the wav2vec 2.0 framework [1]. The raw audio was resampled to 16 kHz, and 768-dimensional features were extracted from the model’s last hidden state. Because the audio and video had different sampling rates, the audio embeddings were temporally resampled to match the video frame count.

Model Architecture: We developed the AudioToBlendshape network, a sequence-to-sequence model to capture the non-linear relationship between phonetics and facial motion, see Figure 1. For the input processing, aligned audio features are projected to a lower dimension (512) and normalized via Layer Normalization. Moreover, a 3-layer Bidirectional LSTM processes the sequence to capture temporal context from both past and future speech segments, outputting a concatenated hidden state of dimension 1024. To handle long-range dependencies and emphasize specific phonetic triggers, we implemented a Multi-Head Self-Attention block (8 heads) with residual connections and layer normalization. Lastly, the output passes through a feed-forward network with ReLU activation and Dropout ($p=0.5$) before being projected to the final 52 blendshape values. We optimized the network using the AdamW optimizer ($\beta_1 = 0.0003$, weight decay=0.0001) with Mean Squared Error (MSE) as the loss function. To ensure training stability and generalization, we employed a ReduceLROnPlateau scheduler to adjust the learning rate upon validation stagnation, alongside an Early Stopping mechanism. Gradient clipping was applied to prevent exploding gradients during backpropagation.

Deployment and Visualization: Following the training stage, we deployed the model into an XR environment. The trained model was exported to the ONNX format and integrated into the Unity Game Engine (v6000.0.42f1) utilizing Unity Senti (v2.1.3). We implemented the XR features using the Meta XR SDKs (v81) for the Quest 3 headset.

4 RESULTS & DISCUSSION

The model achieved an MSE of 0.0088 and an RMSE of 0.0939. We analyzed the per-blendshape error distribution to identify specific areas of difficulty. The model achieved its highest accuracy on blendshapes indices [6, 8, 23, 0, 7], which typically correspond to broad, frequent facial movements. Conversely, the highest error rates were observed in indices [3, 4, 38, 49, 5]. This quantitative data aligns with our visual observations, particularly regarding the subtle dynamics of the upper face.

A key observation was the model’s sensitivity to emotional intensity, see Figure 2. The system demonstrated superior performance when synthesizing level 3 (high intensity) emotions com-

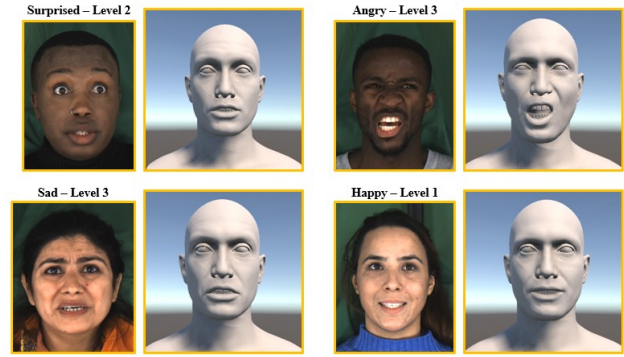


Figure 2: Visual inspection of the generated facial animations.

pared to neutral or low-intensity inputs. Despite the low MSE, visual inspection revealed limitations in the synthesis of the upper face. Specifically, the model struggled to accurately reconstruct the distance and tension between the eyebrows. In future research, we aim to train on the full MEAD dataset to improve the resolution of subtle expressions and brow dynamics. The current blendshape standard controls surface geometry but does not inherently manage internal anatomical constraints. During testing, we noted that while the jawOpen blendshape deformed the skin, it did not strictly decouple the upper and lower teeth separation. We addressed this via a hardcoded procedural offset for the teeth meshes in Unity. A fully learned solution could be explored for explicit skeletal bones or distinct blendshapes for dental structures. Moreover, wav2vec is computationally expensive for XR devices. In future research, we will investigate distilled acoustic models to reduce the computational cost of feature extraction without sacrificing expressiveness. Finally, this work represents one stage of a broader envisioned framework. Future iterations will aim to integrate this module with a generative personalized mesh system (to customize the 3D face mesh because the current pipeline maps outputs to a fixed template mesh) and a generative audio module. This would enable a complete, autonomous system where an agent generates its own emotionally annotated speech and corresponding facial animation in real-time.

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